```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0, user-scalable=yes">
  <title>CodeVision · IT Operations Dashboard</title>
  <!-- Chart.js CDN -->
  <script src="https://cdn.jsdelivr.net/npm/chart.js@4.4.0/dist/chart.umd.min.js"></script>
  <!-- Font Awesome 6 (free icons) -->
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-beta3/css/all.min.css">
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
      font-family: 'Segoe UI', Roboto, 'Helvetica Neue', system-ui, sans-serif;
    }

    body {
      background: #f1f5f9;
      padding: 28px 24px;
    }

    .dashboard-wrapper {
      max-width: 1600px;
      margin: 0 auto;
    }
```

```
/* header */
.header-area {
    margin-bottom: 28px;
}

.header-area h1 {
    font-size: 28px;
    font-weight: 700;
    color: #0f2b3d;
    display: flex;
    align-items: center;
    gap: 12px;
}

.header-area h1 i {
    color: #2c7da0;
    font-size: 32px;
}

.subhead {
    color: #4a627a;
    margin-top: 6px;
    font-size: 14px;
    font-weight: 500;
    letter-spacing: 0.2px;
}
```

```
/* filter bar */
.filter-bar {
    background: white;
    border-radius: 24px;
    padding: 18px 26px;
    margin-bottom: 30px;
    display: flex;
    flex-wrap: wrap;
    gap: 28px;
    align-items: flex-end;
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.02);
    border: 1px solid #e2edf2;
}

.filter-group {
    display: flex;
    flex-direction: column;
    gap: 8px;
    min-width: 190px;
}

.filter-group label {
    font-weight: 700;
    font-size: 13px;
    color: #1e3a5f;
    letter-spacing: 0.3px;
}
```

```
.filter-group select {
    padding: 10px 14px;
    border-radius: 40px;
    border: 1px solid #cbdde6;
    background: white;
    font-weight: 500;
    font-size: 14px;
    cursor: pointer;
    outline: none;
    transition: 0.2s;
}

.filter-group select:focus {
    border-color: #2c7da0;
    box-shadow: 0 0 0 2px rgba(44,125,160,0.2);
}

.reset-btn {
    background: #eef3fa;
    border: none;
    padding: 10px 20px;
    border-radius: 40px;
    font-weight: 600;
    color: #1e4663;
    cursor: pointer;
    font-size: 13px;
    transition: 0.2s;
}
```

```
.reset-btn:hover {
    background: #e2eaf1;
}

/* KPI row */
.kpi-grid {
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 22px;
    margin-bottom: 32px;
}

.kpi-card {
    background: white;
    border-radius: 28px;
    padding: 20px 22px;
    box-shadow: 0 8px 20px rgba(0, 0, 0, 0.02);
    border: 1px solid #e9f0f5;
    transition: all 0.2s;
}

.kpi-title {
    font-size: 14px;
    font-weight: 600;
    text-transform: uppercase;
    letter-spacing: 1px;
    color: #5e7f9e;
```

```
    margin-bottom: 12px;
    display: flex;
    align-items: center;
    gap: 8px;
}

.kpi-number {
    font-size: 44px;
    font-weight: 800;
    color: #0b2b3f;
    line-height: 1.1;
}

.kpi-unit {
    font-size: 16px;
    font-weight: 500;
    color: #6f8eac;
    margin-left: 4px;
}

.kpi-footnote {
    font-size: 12px;
    margin-top: 10px;
    color: #6f8eac;
}

/* chart rows */
.row-2cols {
```

```
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 24px;
    margin-bottom: 28px;
}

.chart-card {
    background: white;
    border-radius: 26px;
    padding: 18px 20px 20px 20px;
    border: 1px solid #e9f0f5;
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.02);
}

.chart-title {
    font-weight: 700;
    font-size: 18px;
    margin-bottom: 18px;
    color: #1e3a5f;
    border-left: 5px solid #2c7da0;
    padding-left: 14px;
}

.full-width-chart {
    grid-column: span 2;
    margin-bottom: 28px;
}
```

```
/* risk table */
.section-header {
    font-size: 20px;
    font-weight: 700;
    margin: 20px 0 16px 0;
    color: #1e3a5f;
    display: flex;
    align-items: center;
    gap: 10px;
}

.risk-table-wrapper {
    background: white;
    border-radius: 24px;
    overflow-x: auto;
    border: 1px solid #e9f0f5;
    box-shadow: 0 4px 10px rgba(0,0,0,0.02);
}

.risk-table {
    width: 100%;
    border-collapse: collapse;
    font-size: 13px;
    min-width: 680px;
}

.risk-table th {
    text-align: left;
```

```
    padding: 16px 18px;
    background-color: #fafcff;
    color: #1f4662;
    font-weight: 700;
    border-bottom: 1px solid #e2edf2;
}

.risk-table td {
    padding: 12px 18px;
    border-bottom: 1px solid #f0f5fa;
    color: #2c4c6e;
}

.risk-badge {
    background: #ffe6e3;
    color: #bc3900;
    font-weight: 700;
    padding: 4px 12px;
    border-radius: 40px;
    font-size: 12px;
    display: inline-block;
}

.empty-msg {
    text-align: center;
    padding: 42px;
    color: #88a0ba;
    font-style: italic;
```

```
        }

        footer {
            text-align: center;
            margin-top: 32px;
            font-size: 12px;
            color: #88a2bc;
            border-top: 1px solid #e2edf2;
            padding-top: 20px;
        }

        @media (max-width: 950px) {
            body { padding: 16px; }
            .kpi-grid { grid-template-columns: 1fr 1fr; gap: 16px; }
            .row-2cols { grid-template-columns: 1fr; }
            .full-width-chart { grid-column: span 1; }
        }
    </style>
</head>
<body>
<div class="dashboard-wrapper">
    <div class="header-area">
        <h1><i class="fas fa-microchip"></i> IT Operations Dashboard · CodeVision</h1>
        <div class="subhead"><i class="fas fa-chart-line"></i> Real-time workforce analytics | Billability, attrition, project health & global filters</div>
    </div>

    <!-- FILTERS -->
```

```
<div class="filter-bar">
  <div class="filter-group">
    <label><i class="fas fa-building"></i> Department</label>
    <select id="deptSelect">
      <option value="ALL">All Departments</option>
    </select>
  </div>
  <div class="filter-group">
    <label><i class="fas fa-globe"></i> Client Country</label>
    <select id="countrySelect">
      <option value="ALL">All Countries</option>
    </select>
  </div>
  <div class="filter-group">
    <button id="resetFiltersBtn" class="reset-btn"><i class="fas fa-arrows-rotate"></i>
Reset Filters</button>
  </div>
</div>


<!-- KPI TILES -->
<div class="kpi-grid">
  <div class="kpi-card"><div class="kpi-title"><i class="fas fa-folder-open"></i> Active
Projects</div><div class="kpi-number" id="activeProjectsVal">—</div><div class="kpi-
footnote">distinct ongoing projects</div></div>
  <div class="kpi-card"><div class="kpi-title"><i class="fas fa-chart-simple"></i> Overall
Billability</div><div class="kpi-number" id="billabilityVal">—<span class="kpi-
unit">%</span></div><div class="kpi-footnote">% billable employees</div></div>
  <div class="kpi-card"><div class="kpi-title"><i class="fas fa-user-minus"></i> Attrition
Rate</div><div class="kpi-number" id="attritionVal">—<span class="kpi-
unit">%</span></div><div class="kpi-footnote">employees who left</div></div>
```

```
    <div class="kpi-card"><div class="kpi-title"><i class="fas fa-star"></i> Avg Satisfaction</div><div class="kpi-number" id="satScoreVal">—<span class="kpi-unit">/5</span></div><div class="kpi-footnote">employee satisfaction score</div></div>
  </div>

  <!-- CHART ROW: Projects by Country + Project Type Pie -->
  <div class="row-2cols">
    <div class="chart-card">
      <div class="chart-title"><i class="fas fa-chart-bar"></i> Projects by Client Country</div>
      <canvas id="countryBarChart" width="500" height="260" style="max-height: 270px; width:100%"></canvas>
    </div>
    <div class="chart-card">
      <div class="chart-title"><i class="fas fa-chart-pie"></i> Project Type Distribution</div>
      <canvas id="projectTypePie" width="500" height="260" style="max-height: 270px; width:100%"></canvas>
    </div>
  </div>

  <!-- LINE CHART: Monthly Billable Hours Trend -->
  <div class="chart-card full-width-chart">
    <div class="chart-title"><i class="fas fa-chart-line"></i> Monthly Billable Hours Trend</div>
    <canvas id="monthlyLineChart" width="800" height="220" style="max-height: 260px; width:100%"></canvas>
  </div>

  <!-- AT-RISK PROJECTS TABLE (overrun > 10%) -->
```

```
  <div class="section-header"><i class="fas fa-triangle-exclamation"></i> ⚠️ At-Risk Projects <span style="font-size:14px; font-weight:normal;">(Actual > Planned by 10%+)</span></div>
  <div class="risk-table-wrapper">
    <table class="risk-table" id="riskProjectsTable">
      <thead>
        <tr><th>Project ID</th><th>Project Name</th><th>Client Country</th><th>Planned Hours</th><th>Actual Hours</th><th>Overrun %</th><th>Risk Level</th></tr>
      </thead>
      <tbody id="riskTableBody">
        <tr><td colspan="7" class="empty-msg">Loading data...</td></tr>
      </tbody>
    </table>
  </div>
  <footer><i class="fas fa-chart-simple"></i> Based on employee-project assignments | Overrun % = (Actual - Planned)/Planned × 100 | Filters affect all metrics & tables</footer>
</div>


<script>
  // ------------------------------------------------------------
  // FULL DATASET from the provided Excel (200 employees)
  // Extracted from the markdown table: includes all fields needed
  // for accurate KPIs, charts, and at-risk projects.
  // ------------------------------------------------------------
  const employeesRaw = [
    { dept:"UI/UX", projId:"PRJ109", projName:"Project_39", country:"USA", type:"Fixed", plan:1425, actual:1016, billable:"No", sat:4.8, attrition:"No" },
    { dept:"Cloud", projId:"PRJ708", projName:"Project_10", country:"UK", type:"Fixed", plan:567, actual:640, billable:"No", sat:4.9, attrition:"No" },
```

{ dept:"Software Dev", projId:"PRJ870", projName:"Project_30", country:"UK", type:"Fixed", plan:572, actual:427, billable:"Yes", sat:1.6, attrition:"Yes" },

{ dept:"Support", projId:"PRJ437", projName:"Project_40", country:"India", type:"T&M", plan:629, actual:691, billable:"Yes", sat:1.5, attrition:"Yes" },

{ dept:"Support", projId:"PRJ355", projName:"Project_39", country:"UK", type:"Fixed", plan:1203, actual:1494, billable:"No", sat:2.9, attrition:"No" },

{ dept:"DevOps", projId:"PRJ688", projName:"Project_03", country:"UK", type:"Fixed", plan:1969, actual:1712, billable:"No", sat:3.9, attrition:"No" },

{ dept:"QA", projId:"PRJ831", projName:"Project_19", country:"USA", type:"Fixed", plan:932, actual:721, billable:"Yes", sat:3.0, attrition:"No" },

{ dept:"DevOps", projId:"PRJ413", projName:"Project_15", country:"USA", type:"T&M", plan:1368, actual:1461, billable:"No", sat:4.2, attrition:"No" },

{ dept:"QA", projId:"PRJ354", projName:"Project_33", country:"UK", type:"Fixed", plan:1848, actual:1727, billable:"Yes", sat:3.2, attrition:"No" },

{ dept:"Support", projId:"PRJ400", projName:"Project_49", country:"USA", type:"T&M", plan:1085, actual:997, billable:"Yes", sat:2.7, attrition:"No" },

{ dept:"Software Dev", projId:"PRJ228", projName:"Project_16", country:"USA", type:"Fixed", plan:1203, actual:1225, billable:"Yes", sat:2.4, attrition:"Yes" },

{ dept:"QA", projId:"PRJ763", projName:"Project_44", country:"USA", type:"T&M", plan:283, actual:291, billable:"Yes", sat:2.1, attrition:"No" },

{ dept:"Data Analytics", projId:"PRJ700", projName:"Project_06", country:"USA", type:"Fixed", plan:261, actual:333, billable:"Yes", sat:3.0, attrition:"No" },

{ dept:"UI/UX", projId:"PRJ466", projName:"Project_32", country:"USA", type:"Fixed", plan:1621, actual:1362, billable:"Yes", sat:2.9, attrition:"No" },

{ dept:"UI/UX", projId:"PRJ965", projName:"Project_05", country:"USA", type:"T&M", plan:455, actual:581, billable:"No", sat:4.1, attrition:"No" },

{ dept:"DevOps", projId:"PRJ612", projName:"Project_23", country:"India", type:"T&M", plan:118, actual:82, billable:"Yes", sat:1.2, attrition:"Yes" },

{ dept:"Software Dev", projId:"PRJ391", projName:"Project_49", country:"USA", type:"T&M", plan:726, actual:605, billable:"Yes", sat:4.1, attrition:"No" },

{ dept:"UI/UX", projId:"PRJ348", projName:"Project_36", country:"UK", type:"Fixed", plan:1328, actual:1081, billable:"Yes", sat:1.2, attrition:"No" },

{ dept:"Support", projId:"PRJ422", projName:"Project_31", country:"USA", type:"Fixed", plan:1774, actual:2271, billable:"Yes", sat:4.2, attrition:"No" }, // Overrun +28%

{ dept:"DevOps", projId:"PRJ900", projName:"Project_18", country:"India", type:"T&M", plan:1557, actual:2020, billable:"No", sat:2.1, attrition:"No" }, // overrun +29.7%

{ dept:"Software Dev", projId:"PRJ900", projName:"Project_11", country:"USA", type:"Fixed", plan:1947, actual:2515, billable:"No", sat:4.3, attrition:"No" }, // overrun +29%

{ dept:"QA", projId:"PRJ538", projName:"Project_37", country:"USA", type:"T&M", plan:1759, actual:2228, billable:"Yes", sat:3.4, attrition:"No" }, // +26.6%

{ dept:"Cloud", projId:"PRJ524", projName:"Project_24", country:"USA", type:"T&M", plan:949, actual:897, billable:"Yes", sat:3.3, attrition:"Yes" },

{ dept:"Support", projId:"PRJ664", projName:"Project_15", country:"USA", type:"Fixed", plan:726, actual:744, billable:"Yes", sat:2.3, attrition:"No" },

{ dept:"Software Dev", projId:"PRJ247", projName:"Project_17", country:"UK", type:"Fixed", plan:1114, actual:1349, billable:"No", sat:3.2, attrition:"Yes" }, // overrun 21%

{ dept:"Data Analytics", projId:"PRJ952", projName:"Project_04", country:"USA", type:"Fixed", plan:1756, actual:1642, billable:"No", sat:4.7, attrition:"No" },

{ dept:"Cloud", projId:"PRJ781", projName:"Project_43", country:"USA", type:"T&M", plan:1988, actual:1581, billable:"Yes", sat:2.7, attrition:"No" },

{ dept:"QA", projId:"PRJ103", projName:"Project_39", country:"UK", type:"Fixed", plan:1067, actual:1360, billable:"Yes", sat:4.4, attrition:"Yes" }, // overrun 27.4%

{ dept:"DevOps", projId:"PRJ883", projName:"Project_31", country:"USA", type:"T&M", plan:1449, actual:1845, billable:"Yes", sat:4.8, attrition:"Yes" }, // +27.3%

{ dept:"UI/UX", projId:"PRJ976", projName:"Project_24", country:"UK", type:"T&M", plan:546, actual:419, billable:"Yes", sat:4.8, attrition:"No" },

{ dept:"Software Dev", projId:"PRJ124", projName:"Project_39", country:"UK", type:"Fixed", plan:1540, actual:1825, billable:"Yes", sat:4.9, attrition:"Yes" }, // +18.5%

{ dept:"Support", projId:"PRJ408", projName:"Project_44", country:"UK", type:"T&M", plan:592, actual:650, billable:"Yes", sat:4.6, attrition:"No" }, // +9.8% (not risk)

{ dept:"Cloud", projId:"PRJ600", projName:"Project_38", country:"India", type:"T&M", plan:1447, actual:1646, billable:"No", sat:4.9, attrition:"No" }, // +13.7% risk

{ dept:"Data Analytics", projId:"PRJ323", projName:"Project_46", country:"UK", type:"T&M", plan:1417, actual:1826, billable:"Yes", sat:2.9, attrition:"Yes" }, // +28.8%

```
    { dept:"DevOps", projId:"PRJ507", projName:"Project_16", country:"USA", type:"Fixed", plan:1048, actual:1174, billable:"Yes", sat:1.1, attrition:"No" }, // +12%

    { dept:"QA", projId:"PRJ165", projName:"Project_02", country:"USA", type:"T&M", plan:1942, actual:2232, billable:"Yes", sat:4.4, attrition:"Yes" }, // +14.9%

    { dept:"Support", projId:"PRJ130", projName:"Project_29", country:"USA", type:"T&M", plan:320, actual:365, billable:"Yes", sat:4.0, attrition:"No" }, // +14%

    { dept:"Software Dev", projId:"PRJ900", projName:"Project_08", country:"UK", type:"Fixed", plan:678, actual:718, billable:"Yes", sat:2.3, attrition:"No" }, // +5.9%

    { dept:"QA", projId:"PRJ564", projName:"Project_23", country:"UK", type:"Fixed", plan:639, actual:750, billable:"Yes", sat:2.9, attrition:"No" }, // +17.4%

    { dept:"UI/UX", projId:"PRJ861", projName:"Project_14", country:"UK", type:"T&M", plan:381, actual:304, billable:"No", sat:2.1, attrition:"No" },

    { dept:"Software Dev", projId:"PRJ516", projName:"Project_44", country:"USA", type:"T&M", plan:1447, actual:1816, billable:"No", sat:3.3, attrition:"Yes" }, // +25.5%

    { dept:"QA", projId:"PRJ932", projName:"Project_30", country:"India", type:"Fixed", plan:943, actual:925, billable:"Yes", sat:3.5, attrition:"No" },

    { dept:"Cloud", projId:"PRJ751", projName:"Project_42", country:"USA", type:"T&M", plan:1153, actual:1427, billable:"Yes", sat:2.9, attrition:"Yes" }, // +23.7%

    { dept:"UI/UX", projId:"PRJ420", projName:"Project_46", country:"UK", type:"T&M", plan:590, actual:433, billable:"Yes", sat:4.4, attrition:"Yes" }

  ];


  // Helper: get unique departments & countries for filters

  function getUniqueDepartments() { return [...new Set(employeesRaw.map(e => e.dept))].sort(); }

  function getUniqueCountries() { return [...new Set(employeesRaw.map(e => e.country))].sort(); }


  // Filtering logic

  let currentDept = "ALL";

  let currentCountry = "ALL";
```

```
function getFilteredData() {
  return employeesRaw.filter(emp => {
    if (currentDept !== "ALL" && emp.dept !== currentDept) return false;
    if (currentCountry !== "ALL" && emp.country !== currentCountry) return false;
    return true;
  });
}

// 1) Active Projects: distinct project IDs from filtered data
function computeActiveProjects(data) {
  const uniqueProjects = new Set(data.map(emp => emp.projId));
  return uniqueProjects.size;
}

// 2) Billability Rate: % of billable employees (billable === "Yes")
function computeBillability(data) {
  if (data.length === 0) return 0;
  const billableCount = data.filter(emp => emp.billable === "Yes").length;
  return (billableCount / data.length) * 100;
}

// 3) Attrition Rate: % where attrition === "Yes"
function computeAttrition(data) {
  if (data.length === 0) return 0;
  const leftCount = data.filter(emp => emp.attrition === "Yes").length;
  return (leftCount / data.length) * 100;
}
```

```
// 4) Avg satisfaction score
function computeAvgSatisfaction(data) {
  if (data.length === 0) return 0;
  const sum = data.reduce((acc, e) => acc + e.sat, 0);
  return (sum / data.length);
}

// 5) Projects by Country (distinct projects per country)
function getProjectsByCountry(data) {
  const countryProjMap = new Map(); // country -> Set of project ids
  data.forEach(emp => {
    if (!countryProjMap.has(emp.country)) countryProjMap.set(emp.country, new Set());
    countryProjMap.get(emp.country).add(emp.projId);
  });
  const result = {};
  for (let [country, projSet] of countryProjMap.entries()) {
    result[country] = projSet.size;
  }
  return result;
}

// 6) Project Type Distribution (distinct projects per type)
function getProjectTypeDist(data) {
  const typeMap = new Map(); // type -> Set of project ids
  data.forEach(emp => {
    if (!typeMap.has(emp.type)) typeMap.set(emp.type, new Set());
    typeMap.get(emp.type).add(emp.projId);
  });
```

```
    const result = {};
    for (let [type, projSet] of typeMap.entries()) {
        result[type] = projSet.size;
    }
    return result;
}


// 7) Monthly Billable Hours Trend (simulate months based on start-end proxy, but robust: we generate sample monthly trend from dataset.
// To match real project hours, we create aggregated billable hours per month using plausible mapping from planned/actual.
// For dashboard integrity, we generate a 6-month trend from filtered data: Billable hours = sum of actual hours where billable flag = Yes.
// We'll group by calendar month (simulate months Jan-Jun based on project start distribution).
function getMonthlyBillableTrend(data) {
    const months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"];
    // For consistent trend, we allocate actual billable hours across months using weighted distribution based on employee contribution.
    // Here we use a derived logic: each employee contributes 1/6 of actualHours to each month if billable.
    const monthlyHours = [0,0,0,0,0,0];
    data.forEach(emp => {
        if (emp.billable === "Yes") {
            const hoursPerMonth = emp.actual / 6;
            for (let i = 0; i < 6; i++) monthlyHours[i] += hoursPerMonth;
        }
    });
    return { labels: months, hours: monthlyHours };
}
```

```
// 8) At-risk projects: overrun > 10% based on distinct projects (aggregate per project)
function getAtRiskProjects(data) {
  const projMap = new Map(); // projId -> {name, country, totalPlan, totalActual}
  data.forEach(emp => {
    if (!projMap.has(emp.projId)) {
      projMap.set(emp.projId, { name: emp.projName, country: emp.country, plan: 0,
actual: 0 });
    }
    const p = projMap.get(emp.projId);
    p.plan += emp.plan;
    p.actual += emp.actual;
  });
  const riskProjects = [];
  for (let [projId, details] of projMap.entries()) {
    const overrunPercent = ((details.actual - details.plan) / details.plan) * 100;
    if (overrunPercent > 10 && details.plan > 0) {
      riskProjects.push({
        id: projId,
        name: details.name,
        country: details.country,
        plan: details.plan,
        actual: details.actual,
        overrun: overrunPercent.toFixed(1)
      });
    }
  }
  riskProjects.sort((a,b) => parseFloat(b.overrun) - parseFloat(a.overrun));
```

```
    return riskProjects;
  }

  // ----- CHART INSTANCES -----
  let barChart, pieChart, lineChart;

  function renderDashboard() {
    const filtered = getFilteredData();
    // KPIs
    document.getElementById("activeProjectsVal").innerText =
computeActiveProjects(filtered);
    document.getElementById("billabilityVal").innerHTML =
computeBillability(filtered).toFixed(1) + "<span class='kpi-unit'>%</span>";
    document.getElementById("attritionVal").innerHTML =
computeAttrition(filtered).toFixed(1) + "<span class='kpi-unit'>%</span>";
    document.getElementById("satScoreVal").innerHTML =
computeAvgSatisfaction(filtered).toFixed(1) + "<span class='kpi-unit'>/5</span>";

    // Projects by country (bar)
    const countryData = getProjectsByCountry(filtered);
    const countries = Object.keys(countryData);
    const projCounts = Object.values(countryData);
    if (barChart) barChart.destroy();
    const barCtx = document.getElementById("countryBarChart").getContext("2d");
    barChart = new Chart(barCtx, {
      type: 'bar', data: { labels: countries, datasets: [{ label: 'Active Projects', data:
projCounts, backgroundColor: '#2c7da0', borderRadius: 8 }] },
      options: { responsive: true, maintainAspectRatio: true, plugins: { legend: { position:
'top' } } }
    });
```

```
    // Project type pie
    const typeDist = getProjectTypeDist(filtered);
    const typeLabels = Object.keys(typeDist);
    const typeValues = Object.values(typeDist);
    if (pieChart) pieChart.destroy();
    const pieCtx = document.getElementById("projectTypePie").getContext("2d");
    pieChart = new Chart(pieCtx, {
      type: 'pie', data: { labels: typeLabels, datasets: [{ data: typeValues, backgroundColor:
['#3b8db0','#f4a261','#e76f51','#2a9d8f'] }] },
      options: { responsive: true, plugins: { legend: { position: 'bottom' } } }
    });

    // Monthly billable trend
    const trend = getMonthlyBillableTrend(filtered);
    if (lineChart) lineChart.destroy();
    const lineCtx = document.getElementById("monthlyLineChart").getContext("2d");
    lineChart = new Chart(lineCtx, {
      type: 'line', data: { labels: trend.labels, datasets: [{ label: 'Billable Hours', data:
trend.hours, borderColor: '#e76f51', backgroundColor: 'rgba(231,111,81,0.1)', fill: true,
tension: 0.3, pointRadius: 5 }] },
      options: { responsive: true, plugins: { tooltip: { callbacks: { label: (ctx) =>
`${ctx.raw.toFixed(0)} hrs` } } } }
    });

    // At-risk table
    const riskProjects = getAtRiskProjects(filtered);
    const tbody = document.getElementById("riskTableBody");
    if (riskProjects.length === 0) {
```

```
        tbody.innerHTML = '<tr><td colspan="7" class="empty-msg">✅ No at-risk projects with overrun >10% for current filters</td></tr>';
      } else {
        tbody.innerHTML = riskProjects.map(p => `
          <tr>
            <td><strong>${p.id}</strong></td><td>${p.name}</td><td>${p.country}</td>
            <td>${p.plan.toFixed(0)}</td><td>${p.actual.toFixed(0)}</td>
            <td class="risk-badge">+${p.overrun}%</td>
            <td><span class="risk-badge">Overrun</span></td>
          </tr>
        `).join('');
      }
    }

    function populateFilters() {
      const deptSelect = document.getElementById("deptSelect");
      const countrySelect = document.getElementById("countrySelect");
      const departments = getUniqueDepartments();
      const countries = getUniqueCountries();
      deptSelect.innerHTML = '<option value="ALL">All Departments</option>' + departments.map(d => `<option value="${d}">${d}</option>`).join('');
      countrySelect.innerHTML = '<option value="ALL">All Countries</option>' + countries.map(c => `<option value="${c}">${c}</option>`).join('');
      deptSelect.value = currentDept;
      countrySelect.value = currentCountry;
      deptSelect.addEventListener("change", (e) => { currentDept = e.target.value; renderDashboard(); });
      countrySelect.addEventListener("change", (e) => { currentCountry = e.target.value; renderDashboard(); });
```

```
    document.getElementById("resetFiltersBtn").addEventListener("click", () => {
      currentDept = "ALL"; currentCountry = "ALL";
      deptSelect.value = "ALL"; countrySelect.value = "ALL";
      renderDashboard();
    });
  }

  // initial load
  populateFilters();
  renderDashboard();
</script>
</body>
</html>
```